Instructor Performance and Course Quality Evaluation on EduPro: A Data-Driven Approach to Educational Quality Assurance

Abstract

The increasing adoption of digital learning platforms has created opportunities to evaluate instructional performance and course quality using data analytics. This study examines instructor ratings, course ratings, teaching experience, areas of expertise, and transaction activity on EduPro, an online education platform. The project dataset comprises 60 teachers, 60 courses, and 10,000 transaction records. Reported descriptive statistics indicate an average teacher rating of 3.125 out of 5, an average course rating of 3.098 out of 5, and average teaching experience of 6.28 years. Correlation analysis indicates a positive association between teaching experience and teacher ratings (Pearson's $r = 0.598$; Spearman's $\rho = 0.671$). In contrast, the association between teaching experience and course ratings is weak and statistically insignificant (Pearson's $r = -0.138$, $p \approx 0.293$). These findings suggest that instructor evaluations and course quality assessments should be treated as related but distinct dimensions of educational performance. The study proposes a data-driven framework for quality assurance, instructor development, course improvement, and evidence-informed decision-making. However, the results are observational and do not establish causal relationships. Further research using raw data, learner-outcome measures, and adjusted statistical models is recommended.

Keywords: Educational analytics, instructor performance, course quality, teaching experience, correlation analysis, quality assurance, online learning.

1. Introduction

Digital education platforms generate substantial quantities of data about instructors, courses, learner interactions, ratings, and transactions. Appropriate analysis of these data can help educational institutions identify performance patterns, evaluate course quality, and improve learning experiences.

Traditional evaluation approaches may rely heavily on subjective feedback or isolated performance indicators. Although learner ratings can provide useful information, they may not fully represent teaching effectiveness, course design, or learning outcomes. A systematic analytical framework is therefore needed to interpret different indicators together while recognizing their limitations.

The EduPro Instructor Performance and Course Quality Evaluation project addresses this need by integrating instructor information, course information, and transaction records into a unified analytical workflow. The project uses Python-based data processing, statistical analysis, visualizations, and a Streamlit dashboard to support transparent monitoring.

The study focuses on the relationship between teaching experience and teacher ratings, the relationship between experience and course ratings, and the use of descriptive indicators for educational quality assurance.

2. Problem Statement

Educational platforms need reliable methods to distinguish instructor performance from course quality. A highly experienced instructor may receive strong teacher ratings without necessarily being associated with

highly rated courses. Similarly, transaction volume may reflect demand without directly measuring educational effectiveness.

Without a structured analytical framework, decision-makers may overinterpret ratings, overlook differences between course categories, or make comparisons without considering sample sizes and contextual factors.

This study seeks to establish a data-driven baseline for evaluating instructor and course performance on EduPro while identifying the methodological limitations that should be addressed before the findings are used for high-impact administrative decisions.

3. Research Objectives

The primary objective is to develop and apply a data-driven framework for evaluating instructor performance and course quality on EduPro.

The specific objectives are:

1. To summarize instructor ratings, course ratings, teaching experience, and transaction activity.
2. To examine the association between teaching experience and teacher ratings.
3. To investigate the association between teaching experience and course ratings.
4. To compare performance indicators across instructor expertise areas and course categories.
5. To develop a dashboard for monitoring key performance indicators and data quality.

6. To recommend practical measures for improving instructional quality and course design.

4. Research Questions

RQ1: What are the descriptive characteristics of instructor performance and course quality on EduPro?

RQ2: Is teaching experience associated with teacher ratings?

RQ3: Is teaching experience associated with course ratings?

RQ4: How can educational analytics support quality assurance and evidence-informed decision-making?

5. Research Hypotheses

H1: Teaching experience is significantly associated with teacher ratings.

H2: Teaching experience is significantly associated with course ratings.

The first hypothesis is supported by the reported correlation results. The second hypothesis is not supported by the reported Pearson correlation analysis at the conventional 5% significance level.

6. Literature and Conceptual Background

Educational quality is multidimensional. It includes instructor knowledge, pedagogical practice, learner support, curriculum relevance, assessment design, accessibility, and learning outcomes. Consequently, a single rating cannot fully capture educational effectiveness.

Instructor evaluations commonly use learner feedback, peer observations, professional qualifications, and evidence of teaching

practice. Learner ratings are useful indicators of perceived teaching quality, but they can be influenced by expectations, course difficulty, response bias, and the number of ratings received.

Course quality is distinct from instructor performance. It can depend on the structure of learning materials, alignment between objectives and assessments, content accuracy, usability, accessibility, and relevance to learners. A course may receive a lower rating because of its difficulty or design even when the associated instructor is highly regarded.

Educational analytics provides methods for combining operational data with performance indicators to identify patterns and support improvement. However, analytical outputs should be interpreted in context. Correlation indicates statistical association rather than causation, and differences between groups may reflect factors other than the performance characteristic being studied.

This study applies these principles by analyzing instructor ratings and course ratings separately and by treating transaction activity as an operational indicator rather than a direct measure of learning effectiveness.

7. Research Methodology

7.1 Research Design

The study adopts a quantitative, observational, and descriptive-correlational research design. It uses existing platform records rather than an experimental intervention.

7.2 Dataset Description

The project dataset is reported to contain the following records:

Dataset	Number of records	Main variables
Teachers	60	Teacher ID, name, rating, experience, expertise
Courses	60	Course ID, name, category, level, rating
Transactions	10,000	Transaction ID, teacher ID, course ID; amount if available

Teacher ratings and course ratings are measured on a five-point scale. Teaching experience is measured in years. Transaction records connect teacher and course identifiers.

These figures represent the project's reported dataset. The final submitted paper should verify them against the actual CSV files and document the dataset extraction date.

7.3 Data Preprocessing

The analytical workflow includes:

1. Loading the teacher, course, and transaction datasets.
2. Checking the presence of required columns.
3. Converting numeric variables to appropriate numeric formats.
4. Removing duplicate teacher, course, and transaction identifiers.
5. Joining transaction records to the corresponding teacher and course records.
6. Checking unmatched identifiers and referential integrity.

7. Generating summary tables and visualizations.

The reported project run indicates valid joins and no unmatched join keys. This establishes a degree of structural consistency but does not guarantee that all variables are complete, unbiased, or accurately measured.

7.4 Statistical Techniques

Descriptive statistics: Means and other summary measures are used to describe teacher ratings, course ratings, and teaching experience.

Pearson correlation: Pearson's correlation coefficient measures the direction and strength of a linear association between two numeric variables.

Spearman rank correlation: Spearman's correlation coefficient measures the strength and direction of a monotonic relationship based on ranked observations.

Data visualization: Scatter plots, bar charts, and category-by-level heatmaps help identify patterns across instructors and courses.

Dashboard reporting: A Streamlit dashboard presents key performance indicators, filters, statistical results, and data-quality checks.

7.5 Significance Level

The conventional significance level is set at $\alpha = 0.05$. Results with p-values below 0.05 are treated as statistically significant, subject to the assumptions and limitations of the analyses.

The p-values reported below are approximate calculations based on the reported correlation coefficients and a sample size of 60. Exact values

should be verified using the unrounded coefficients and raw data before formal publication.

8. Results and Analysis

8.1 Dataset Overview

The project dataset contains 60 teachers, 60 courses, and 10,000 transactions. The transaction table provides an operational record of platform activity, while teacher and course tables provide information about instructional personnel and learning products.

The reported data-quality checks indicate that teacher and course identifiers could be joined to transaction records without unmatched keys. This supports the use of the combined dataset for the reported descriptive analysis.

8.2 Descriptive Findings

Indicator	Reported result
Number of teachers	60
Number of courses	60
Number of transactions	10,000
Average teacher rating	3.125/5
Average course rating	3.098/5
Average teaching experience	6.28 years

The average teacher rating and average course rating are both approximately 3.1 out of 5. These averages provide an initial

performance baseline. However, they do not independently establish whether educational quality is satisfactory.

Further interpretation requires rating distributions, rating counts, course-category comparisons, and learning-outcome measures.

8.3 Teaching Experience and Teacher Ratings

The reported Pearson correlation between teaching experience and teacher rating is **r = 0.598**. This indicates a moderately strong positive linear association.

The reported Spearman correlation is **ρ = 0.671**, indicating a positive monotonic relationship between the two variables.

Using the reported coefficients and n = 60, the approximate two-sided p-values are:

- Pearson correlation: $p \approx 4.5 \times 10^{-7}$.
- Spearman correlation: $p \approx 4.4 \times 10^{-9}$.

Both results are statistically significant at the 5% level.

These findings suggest that, within the observed dataset, teachers with greater experience tend to receive higher teacher ratings. However, this relationship does not demonstrate that experience itself causes higher ratings. Other factors, including subject expertise, communication, learner expectations, course assignment, and rating counts, may contribute to the observed pattern.

8.4 Teaching Experience and Course Ratings

The reported Pearson correlation between teaching experience and associated course ratings is **r = −0.138**, with an approximate p-value of **0.293**.

The relationship is weak and is not statistically significant at $\alpha = 0.05$. Therefore, the available analysis does not provide sufficient evidence of a linear association between teaching experience and course ratings.

This finding highlights the importance of assessing course quality independently of instructor tenure. Course ratings may be influenced by course structure, content, difficulty, assessments, accessibility, and learner expectations.

An important limitation is that course ratings are course-level measures. When a course is associated with multiple teachers, its rating cannot automatically be interpreted as an exclusive measure of any one teacher's contribution.

8.5 Transaction Activity

The dataset includes 10,000 transactions. This provides a substantial record of operational activity for descriptive monitoring.

Nevertheless, transaction count should be interpreted as enrollment or demand only after confirming what each transaction represents. If transactions include payments, renewals, refunds, or multiple purchases by the same learner, transaction volume may not equal the number of unique enrollments.

Demand indicators should not be treated as direct evidence of learning gains or instructional effectiveness without additional measures such as completion rates, assessment performance, and learner progress.

9. Discussion

The main finding is that teaching experience is positively associated with teacher ratings, while no statistically significant linear relationship was observed between experience and course ratings.

These findings suggest that perceived instructor performance and course quality may represent different dimensions of the learning experience. Teacher ratings may reflect learner perceptions of communication, responsiveness, subject knowledge, or instructional delivery. Course ratings may reflect content design, organization, difficulty, assessment alignment, and the overall learning experience.

The stronger Spearman coefficient relative to Pearson's coefficient suggests that the positive ordering of teacher experience and ratings may be clearer in the ranked data than in a strictly linear model. However, the analysis does not establish the mechanisms behind the relationship.

The descriptive averages of approximately 3.1/5 also indicate that aggregate means should be supplemented with more detailed analysis. Administrators should examine the distribution of ratings, the number of responses per teacher and course, and differences across subject categories and course levels.

The findings support a balanced quality-assurance approach that uses quantitative indicators to identify areas for further investigation rather than treating them as definitive judgments of educational effectiveness.

10. Recommendations

10.1 Instructor Development

Develop mentoring and peer-learning opportunities that allow instructors to share effective teaching practices. Use ratings as an initial signal for identifying support needs, not as the sole basis for employment or performance decisions.

10.2 Course Quality Audits

Evaluate course structure, content accuracy, learning objectives, assessment alignment, accessibility, and learner support separately from teacher ratings.

10.3 Improved Data Collection

Record rating counts, course enrollment, completion rates, assessment outcomes, learner feedback, and course assignment dates. These additional variables can improve the validity of future analyses.

10.4 Transparent KPI Definitions

Document the meaning, calculation, limitations, and reporting frequency of each KPI. Distinguish established statistical measures from project-defined indicators.

10.5 Evidence-Informed Governance

Use dashboards as decision-support tools. Require human review, contextual evidence, and adequate sample sizes before making high-impact decisions about instructors or courses.

10.6 Continuous Monitoring

Conduct periodic analyses to identify persistent quality concerns, evaluate interventions, and measure changes in learner satisfaction and learning outcomes.

11. Limitations

The study has several limitations.

First, the analysis contains 60 teacher records and 60 course records. Although the transaction table contains 10,000 records, the teacher-level correlations are based on teacher-level observations rather than 10,000 independent observations.

Second, the analysis is observational and cannot establish causation. Unmeasured factors may influence both teaching experience and ratings.

Third, rating counts and response distributions are not included in the reported summary statistics. Unequal numbers of ratings may affect the reliability and comparability of averages.

Fourth, course ratings cannot be attributed exclusively to an individual instructor when multiple teachers are associated with the same course.

Fifth, transaction activity may not represent unique enrollments or educational outcomes.

Finally, the reported summary statistics have not been independently recalculated from the raw CSV files for this draft. Exact confidence intervals, adjusted models, and subgroup analyses should be included in a final empirical version.

12. Conclusion

This study presents a data-driven framework for evaluating instructor performance and course quality on EduPro. The reported findings indicate a positive association between teaching experience and teacher ratings, but no statistically significant linear association between teaching experience and course ratings.

The results demonstrate why instructor evaluations and course-quality assessments should be considered related but distinct dimensions of educational performance. Teaching experience may be associated with perceived instructor performance, but it should not be used as a substitute for direct course-quality evaluation.

The project's data-processing workflow and dashboard can support ongoing monitoring, provided that metrics are clearly defined and interpreted within their limitations. Future work should validate the reported statistics against raw data, incorporate learning-outcome measures, and evaluate whether targeted interventions lead to measurable improvements.

References

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EduPro project dataset and analytical outputs. Project-provided data and summary statistics. The dataset version, extraction date, and data provenance should be documented before formal submission.

Appendix: KPI Interpretation

- **Average teacher rating:** Mean teacher-level rating on the platform's five-point scale.
- **Average course rating:** Mean course-level rating on the five-point scale.
- **Average teaching experience:** Mean number of years of experience across teacher records.
- **Transaction count:** Number of distinct transaction identifiers associated with a teacher; an enrollment proxy only if supported by the transaction definition.
- **Rating consistency index:** A project-defined indicator based on variation in associated course ratings. It should not be presented as a standardized industry metric.
- **Correlation coefficient:** A measure of association that does not establish causation.